

MULTIPLE UNIT DISPATCHES IN EMERGENCY SERVICES: MODELS TO ESTIMATE SYSTEM PERFORMANCE*

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One emergency service system model that has undergone extensive development and testing is the hypercube queuing model and its approximate analog, which were developed by Larson. The model estimates key system performance measures, such as individual unit workloads and travel times. One limitation of this model, as well as other probabilistic deployment models, is that they assume only a single unit is dispatched to each call. In the police environment, if two officers are assigned to each unit, this assumption is generally valid. However, most fire services dispatch multiple vehicles to a serious fire, and in police departments with one-officer units, a second unit will be sent to all potentially dangerous calls.

The need for a new model goes beyond an interest in more accurately estimating standard performance measures when multiple units are dispatched. New models are required to predict new performance measures that are relevant only when multiple units are dispatched. For example, what are the paired travel times of the first- and second-arriving units at an emergency? How long will the first unit be exposed at a potentially dangerous situation until a backup unit arrives? Which unit most frequently arrives first at calls requiring multiple units?

In this paper we build on Larson's work and present two models, one exact, the other, approximate, which can capture the simultaneous response of two identical units dispatched to a single call. We present an example from the police context to illustrate both models and compare the speed and accuracy of the approximate model to the exact one. The workload and travel time estimates of both models differ on the average by only two percent. We conclude our presentation with a brief discussion of model extensions with a focus on the analysis of a merger of police, fire and emergency medical services into a public safety service. (GOVERNMENT SERVICES—POLICE; PERFORMANCE MEASUREMENT; PROBABILITY—STOCHASTIC MODEL APPLICATIONS)

1. Introduction

In recent years numerous models have been developed to assist administrators in the deployment of emergency services. The general problem these models address is how to distribute emergency service units (police cars, fire companies, ambulances) throughout a study region. For an overview of model development in this area see [3] and [17]. One probabilistic analytic model which has generated a great deal of research is the hypercube queuing model developed by Larson [17]. The main advantages of this model with respect to performance measurement are 1) it explicitly accounts for interdistrict dispatching, 2) it allows emergency units to be on the move (e.g. police cars on patrol), and 3) it calculates average travel times for the study region as a whole and for each district, subdistrict and emergency unit within the region. The basic model was first programmed by Campbell [2] under Larson's guidance but was made truly computationally feasible by Larson's work in developing efficient exact [17] and later approximate [18] versions of the model along with an interactive interface [24]. Jarvis [13] modified the model to allow for multiple call types, differentiated by their

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service time, and incorporated the model into an optimization procedure using a Markovian decision process model. Chelst [5] embedded the model in a system that guides a decision-maker in developing an equitable configuration. Larson and Franck expanded the model's dispatch options [19] to allow for closest unit dispatching; other work has been done in expanding the travel time statistics to include the distribution of travel times [7] and to allow for better modeling of ambulance services [1]. The model has been used in a number of cities [1], [14] and field tested by Stenzel [20].

One key assumption of the hypercube model development to date is that only a single unit is dispatched to a call. This assumption is reasonable in locales that assign two patrol officers to each police car, or that have an ambulance-centered emergency medical system. However, police departments with one-officer units frequently dispatch two units to potentially dangerous calls, and emergency medical systems that combine an ambulance transport system with well-equipped paramedics simultaneously dispatch a paramedic unit and an ambulance. Fire departments typically dispatch both a ladder and engine company to a call and send even more units to a multiple alarm fire. The present structure of the hypercube model cannot handle these emergency environments.

New Performance Measures

In a New Haven study [4], multiple dispatches were accounted for in the hypercube model by increasing the estimated hourly call for service rate by double-counting calls requiring two units. This minor modification may be able to capture the impact of multiple dispatches on unit workloads. It cannot enable the hypercube model to calculate statistics describing a dual response to a single call. We have therefore modified the model and its approximate counterpart to calculate new performance measures that describe the system response to calls *requiring* two identical units. We call such calls "Type 2" calls, and define the following new statistics:

1. the mean travel time to Type 2 calls,
2. the mean travel time of the primary unit to a Type 2 call,
3. the mean travel time of the backup unit to a Type 2 call,
4. the mean travel time of the first-arriving unit,
5. the mean travel time of the second-arriving unit,
6. the mean and the distribution of the delay between the arrival of the first and second units,
7. the fraction of the entire region's Type 2 calls for which unit i is the first to arrive at the scene.

In order to understand the difference between the statistics numbered 2 and 4, we briefly describe a typical police dispatch operation. When a call comes in, the dispatcher notes the location of the call and then refers to a fixed preference dispatch list for that area of the city. This list ranks the patrol units in order of preference to calls in that area of the city. The primary unit dispatched is the highest-ranked *available* unit and the backup unit is the second-highest-ranked available unit. Because the dispatcher does not typically know the exact location of the units, the backup unit may in fact be closer to the call and therefore arrive first. Statistics 2 and 3 describe travel times from a dispatcher's perspective. Statistics 4 and 5 describe the system from the perspective of the citizen caller awaiting the arrival of the two units.

An important concern police officers express with regard to one-officer units is that when they arrive at the scene of a call, they may be exposed to danger for an extended

period of time until a backup unit arrives. Statistic 6 reflects this exposure in terms of the time between the arrival of the first and second units. In addition, because of geographic variations in the system, certain units may tend to arrive first more frequently at Type 2 calls. Statistic 7 captures differences in the frequency with which specific units are exposed to potentially dangerous situations. Our models also calculate the standard workload and travel time statistics that are reported in the basic hypercube model [17].

The models we present have two potential roles.

1. Operational: To help analyze redistricting and deployment in police departments with one-officer units.
2. Policy: To enable performance measures to be compared with the basic hypercube model in order to assess in a particular locale the tradeoff between one- and two-officer units. (For an aggregate level discussion of the differences between one- and two-officer units see [6] and [15].)

2. Introduction to Models and Their Assumptions

In the succeeding sections we present two models that capture the impact of dispatching two identical units to a single call. One model is an exact hypercube queuing type model and the other is an approximate analog. Our discussion of the mathematical structure of these models focuses on those aspects of the models that are different from the exact [17] and approximate [18] models of Larson. We begin with a discussion of the assumptions common to both models and the ramifications of these assumptions. We then proceed to describe two different procedures for calculating three key statistics

1. the workload of each unit,
2. the fraction of dispatches to calls requiring one officer that send unit i to location m ,
3. the fraction of dispatches to calls requiring two officers that send units i and j to location m .

These statistics are then used in the same way in both models to calculate other system performance measures such as those listed earlier. Next, we illustrate both models with police data from New Haven, Connecticut and compare the estimates and the speed of the two models. We conclude our presentation with a discussion of extensions of the models and approaches to relaxing certain of the model assumptions.

Central to the models is a geographical description of the study region. The region is partitioned into M geographical atoms. Associated with each atom are a number of pieces of data:¹

- (1) (x, y) coordinates of each atom and/or a matrix of travel times for every pair of atoms $\|\tau_{nm}\|$,
- (2) l_{im} : the proportion of time unit i spends in atom m while it is available,
- (3) f_{m1} and f_{m2} : the fraction of the entire region's calls that require one unit (Type 1 calls) and two units (Type 2 calls) respectively, and that are generated within atom m .

Type 1 calls arise in the region at a rate of λ_1 , and Type 2 calls at a rate of λ_2 . Each stream of calls forms a Poisson process. Type 1 calls are serviced by a single unit and the service time is exponentially distributed with a mean of $1/\mu$. For Type 2 calls, we

¹ All symbols used in this paper are defined in Appendix I.

assume the service times of the two units are independent and each is exponentially distributed with mean $1/\mu$. As a result of this critical assumption, two units servicing a Type 2 call can be treated the same as two units servicing two separate Type 1 calls. In the hypercube queuing model each server's identity is maintained and with the above assumption the state space contains 2^N states. If the service of Type 2 calls were modeled differently, the system would have to differentiate between servers busy at Type 1 and Type 2 calls and the state space would grow to 3^N in the exact model.

The above assumption implies that one response unit will spend an average of $1/2\mu$ hours on the call while a second spends $3/2\mu$. (These are the mean of the minimum and maximum of two independent identically distributed exponential random variables.) This is not inconsistent with data [21] that show both units do not typically spend the same amount of time on a call. The major weakness of this assumption is that the first-arriving unit is not assumed to spend the longer time.

As a consequence of the above assumptions, if we are not concerned with the identity of the busy servers, the queuing system is an $M/M/N$ system with single and bulk arrivals. In the present version of the model we consider only a *zero-line capacity* system that has $N + 1$ states. Zero-line capacity means that 1) Type 1 or Type 2 calls that arrive while all units are busy are either lost or serviced from outside the region, 2) Type 2 calls that arrive when only one unit is available are dispatched a single unit.² A backup is either not sent or assumed to be dispatched from outside the region or from a special reserve group. Recent developments in queuing theory [10, 11] should facilitate the relaxation of the zero-line capacity assumption.

The approximate model builds upon the $M/M/N/0$ queuing model without identifiable servers. If we keep the independent service time assumption, the approximate model describes the state of the system by the number of busy servers; this requires $N + 1$ states. If instead we assume that the two units spend the same amount of time at a Type 2 call but that the mean is still $1/\mu$, the state space in the approximate model would be characterized by the pair (n_1, n_2) , the number of servers alone at a call and the number of servers working together at a Type 2 call. The number of states is equal to the number of possible combinations of n_1 and n_2 . If the mean service time for a Type 2 call were different from that of a Type 1 call, the state space would be characterized by (n_1, n_2, n_3) , which corresponds to the number of servers busy at a Type 1 call, the number of servers working together at a Type 2 call and the number of servers working alone (or with some unit from outside the region) at a Type 2 call. (Remember, the model assumes one unit will be dispatched to a Type 2 call if that is all that is available.) The number of states is equal to the number of different possible combinations of the triplet. For a six-server region, the state space expands from seven (with the independent service time assumption) to sixteen³ and then to 50 states as we relax the service time assumption. For a 10 patrol unit configuration the corresponding numbers are eleven, 36 and 161 states. We have kept

²Because the average service time for Type 1 and Type 2 calls is the same, a single patrol unit at a Type 2 call can be treated the same as a patrol unit at a Type 1 call.

³There are four ways in which six servers can be busy: (1) all are busy at Type 1 calls, (2) four are busy at Type 1 calls and two are busy at one Type 2 call, (3) two are busy at Type 1 calls and four are busy at two Type 2 calls and (4) all are busy at Type 2 calls. Analogously, five servers can be busy in three ways, four servers in three ways, three servers in two ways, two servers in two ways, one server in one way and no servers in one way for a total of 16 possibilities.

the independent service time assumption in the approximate model, which enables us to assess its accuracy and relative speed by comparing it to the exact model.

In both models the dispatch policy is described by a fixed preference list as follows:

(1) For each atom there is a *single* list that ranks the units in order of the dispatch preference to that atom. This list is independent of the state of the system.

(2) When a Type 1 call arises in an atom, the highest ranked available unit is dispatched.

(3) For Type 2 calls, the two highest-ranked available units are dispatched.

(4) In our example the patrol unit ranked first for each atom is the single unit assigned to patrol that atom. The other rankings on the list were found by determining first the average location of each unit. The distance from this location to each atom was then calculated. For each atom these distances were then ordered to form the rankings from 2 to N . This dispatch policy has been labeled modified MCM by Larson [16].

The *single* list for each atom assumes that all units are considered identical except with regard to their location. These models can appropriately reflect the dispatch of two one-officer units or two fire engines. However, two fixed preference lists would be needed to model the simultaneous dispatch of an engine and a ladder to a fire emergency or the dispatch of a paramedic-sheriff's unit and an ambulance to a medical emergency. The first available unit on each list would be dispatched to a call requiring the two different unit types.

3. Exact Model

In order to introduce server identity into the queuing model, we need some additional notation [17]. We represent the status of the N patrol units as a state i and associate with each state a vector $B_i = \{b_1, b_2, \dots, b_N\}$. If $b_n = 1$, unit n is busy, and if it is equal to 0, the unit is available for dispatch. Each possible state of the system is represented by a unique vector of ones and zeros, which corresponds to a unique point or vertex in R^N . The set C_N of all 2^N states is the set of vertices of the N -dimensional unit hypercube in the positive orthant.

The 'Hamming distance' between two vertices B_i and B_j is written as d_{ij} , and is equal to the number of elements of the set B_i that differ from the corresponding elements in B_j . More simply, the Hamming distance is the 'right angle' distance between two vertices in R^N . In Larson's model, all transitions occur between states that are one Hamming distance away; a single server goes into service ($d_{ij}^+ = 1$) or out of service ($d_{ij}^- = 1$). In our model, however, a Type 2 call results in $d_{ij}^+ = 2$ as long as more than one server is free, since two servers go out of service simultaneously.

We have assumed Poisson input and exponential service times, and can calculate $P\{B_j\}$, the steady state probability of being in state j , by writing out equations of detailed balance.

Let $w\{B_i\}$ equal $\sum_i b_i$, the number of servers busy in state B_i . Then

$$\begin{aligned}
 P\{B_j\}(\lambda_1 + \lambda_2 + w\{B_j\}\mu) = & \sum_{\{B_i \in C_N : d_{ij}^+ = 1\}} P\{B_i\}\lambda_{ij}^{(1)} + \sum_{\{B_i \in C_N : d_{ij}^+ = 2\}} P\{B_i\}\lambda_{ij}^{(2)} \\
 & + \sum_{\{B_i \in C_N : d_{ij}^- = 1\}} P\{B_i\}\mu \quad \text{for } j = 0, 1, \dots, 2^N - 2
 \end{aligned}
 \tag{1}$$

$$P\{B_j\}[w\{B_j\}\mu] = \sum_{\{B_i \in C_N : d_{ij}^+ = 1\}} P\{B_i\}(\lambda_{ij}^{(1)} + \lambda_{ij}^{(2)}) + \sum_{\{B_i \in C_N : d_{ij}^+ = 2\}} P\{B_i\}\lambda_{ij}^{(2)} \quad \text{for } j = 2^N - 1. \quad (2)$$

The left hand side is the rate of leaving state j , which occurs either because of new arrivals or one of $w\{B_j\}$ servers completes service. The right hand side is the rate of entering state j , and occurs whenever (1) a Type 1 call's arrival moves the system from state i to j , (2) a Type 2 call's arrival moves the system from state i to j or (3) a service is completed. The last equation is needed because we are assuming zero queue capacity. The system cannot leave state $j = 2^N - 1$ by an arrival, nor can it enter the state by a departure. In addition, Type 2 calls move the system only one Hamming distance if $w\{B_i\}$ equals $N-1$. To guarantee a probability distribution, we require that the state probabilities sum to one. $P\{B_0\}$ and $P\{B_{2^N-1}\}$ can be calculated from the basic $M/M/N/0$ bulk arrival queuing model, since they represent just the probability that no servers are busy or all servers are busy respectively.

Although compactly written, equations (1) and (2) are neither easily determined nor easily solved. The key problem in setting up the equations is determining the $\lambda_{ij}^{(1)}$ and $\lambda_{ij}^{(2)}$. These are the transition rates between states i and j and they depend mainly upon the dispatch policy.

To calculate their values, we determine for each state i and atom m the following:

- (1) the state j that is entered when a Type 1 call arises in atom m and the system is in state i
- (2) the state j' that is entered⁴ when a Type 2 call arises in atom m and the system is in state i .

The state j is found by first searching B_i to determine the available unit n ($b_n = 0$) that is ranked highest with regard to atom m , and then locating the state j whose vector B_j is identical to B_i except that $b_n = 1$. The state j' is found by searching B_i to determine the two highest-ranked (n and k) available units ($b_n = b_k = 0$) with regard to m , and then locating the state j' whose vector $B_{j'}$ is identical to B_i except that $b_n = b_k = 1$. Once j and j' have been identified, we update our estimates as follows:

$$\lambda_{ij}^{(1)} \leftarrow \lambda_{ij}^{(1)} + \lambda_1 f_{m1},$$

$$\lambda_{ij'}^{(2)} \leftarrow \lambda_{ij'}^{(2)} + \lambda_2 f_{m2}.$$

(See Larson [17] for a discussion of the computational and storage problems in the basic hypercube model; these problems are compounded by having two types of calls.)

Solution

We iterate towards a solution of the 2^N simultaneous equations in a manner similar to point Jacobi iteration [22]. We introduce $\hat{P}_n\{B_j\}$ as the estimate of $P\{B_j\}$ after n

⁴In general j and j' will not be the same state. They will be the same state only if $N - 1$ servers are busy in state j .

iterations. The balance equations are rewritten as

$$\hat{P}_n \{ B_j \} = \left[\sum_{B_i \in C_N : d_{ij}^+ = 1} \hat{P}_{n-1} \{ B_i \} \lambda_{ij}^{(1)} + \sum_{B_i \in C_N : d_{ij}^+ = 2} \hat{P}_{n-1} \{ B_i \} \lambda_{ij}^{(2)} + \sum_{B_i \in C_N : d_{ij}^- = 1} \hat{P}_{n-1} \{ B_i \} \mu \right] / [\lambda_1 + \lambda_2 + \mu W \{ B_j \}]$$

for $j = 1, 2, \dots, 2^N - 2$. (3)

$P \{ B_0 \}$ and $P \{ B_{2^N-1} \}$ are known from the basic queuing model without identifiable servers. To start this iterative procedure, we need initial estimates for $\hat{P}_0 \{ B_i \}$. The simplest choice of initial estimates is to assume that all states within one hyperplane are equally likely.

$$P_0 \{ B_j \} = P(w \{ B_j \}) / \binom{N}{w \{ B_j \}}. \quad (4)$$

The denominator is the number of ways of choosing $w \{ B_j \}$ busy servers out of N servers, and is the number of states within the hyperplane, $H_{w \{ B_j \}}$. (All states in a hyperplane have the same number of busy servers.)

The workload of unit n , ρ_n , is found by summing over all states in the hypercube in which unit n is busy (i.e. $b_n = 1$).

$$\rho_n = \sum_{(i : b_n = 1)} P \{ B_i \}.$$

Additional key statistics we calculate are

q_{nm1} = Fraction of all Type 1 dispatches that send unit n to geographical atom m .

$q_{(n,k)m2}$ = Fraction of all Type 2 dispatches that send units n and k to geographical atom m .

q_{nm2} = Fraction of all Type 2 dispatches that send only unit n to geographical atom m .

The Type 1 statistic is calculated by multiplying the proportion of calls generated in atom m by the sum of the state probabilities for which unit n is the highest ranked available unit.

$$q_{nm1} = f_{m1} \sum_{i \in D_m^n} P \{ B_i \}$$

where

$D_m^n \equiv$ set of all states in which unit n is the highest ranked available unit with respect to atom m .

We determine similarly the Type 2 statistics

$$q_{(n,k)m2} = f_{m2} \sum_{i \in D_m^{n,k}} P \{ B_i \}$$

where

$D_m^{n,k} \equiv$ set of all states in which units n and k are the two highest ranked available units with respect to atom m .

and

$$q_{nm2} = f_{m2} P \{ B_i \} \quad \text{for the state } B_i \text{ in which only } b_n = 0.$$

Because we have a zero line capacity queuing system, we normalize the above statistics

so that

$$\sum_m \sum_n q_{nm1} = 1$$

and

$$\sum_m \sum_{n=k+1}^N \left[q_{nm2} + \sum_{k=1}^{N-1} q_{(n,k)m2} \right] = 1.$$

This is done by dividing each statistic by the probability that all N servers are busy.

4. Approximate Model

The exact model presented above has three major drawbacks: 1) storage must be allocated for a state space that grows exponentially as a function of the number of servers (2^N); 2) 2^N simultaneous linear equations must be solved to calculate state probabilities; 3) the transition matrix, which is on the order of $N \times 2^N$, must either be stored or calculated repeatedly. We present here an approximate model that requires only the solution of N simultaneous *nonlinear* equations and requires neither state probabilities nor transition rates. This model is analogous to Larson's approximate model [18] and differs mainly in that different correction factors are required.

The approximate model treats each unit as if its status were independent of the other units' status, which it is not. A correction factor is then introduced to account for this lack of independence. For identical servers the correction factor is exact, but it produces approximate results in the police environment because of non-homogeneous servers.

We again assume a fixed preference dispatch list for each atom. We define

$U_j \equiv j$ th ranked server is unavailable

$F_j \equiv j$ th ranked server is available ("free").

$P\{U_1 U_2 \dots U_{j-1} F_j\}$ = probability that the j th ranked server will be dispatched to a Type 1 call.

For a Type 2 call, the probability that the j th ranked unit will be the *primary* unit dispatched is the same as for a Type 1 call. We are now also interested in the probability that the j th and k th ranked units are simultaneously dispatched to a Type 2 call.

$$P\{U_1 \dots U_{k-1} F_k U_{k+1} \dots U_{j-1} F_j\}$$

We approximate these expressions as follows

$$P\{U_1 U_2 \dots U_{j-1} F_j\} = Q(N, j-1, 1) \prod_{r=1}^{j-1} \rho_{(r)} (1 - \rho_{(j)}), \quad (6)$$

$$P\{U_1 U_2 \dots U_{k-1} F_k U_{k+1} \dots U_{j-1} F_j\} = Q(N, j-1, 2) \prod_{r \neq k}^{j-1} \rho_{(r)} (1 - \rho_{(k)}) (1 - \rho_{(j)}) \quad (7)$$

where

$\rho_{(r)}$ = fraction of time that the r th preferred unit is unavailable for dispatch.

The correction factors, $Q(N, j-1, 1)$ and $Q(N, j-1, 2)$, for Type 1 and Type 2 calls depend on (1) N : the number of servers, (2) j : the rank of the server of interest, and (3) the regionwide average fraction of time units are unavailable for dispatch.

For a set of identical servers, equations (6) and (7) are exact and reduce respectively to

$$Q(N, j-1, 1)\rho^{j-1}(1-\rho) \quad \text{and} \quad Q(N, j-1, 2)\rho^{j-2}(1-\rho)^2. \quad (8)$$

In addition, for a set of identical servers we can calculate directly the probabilities. To find $P\{U_1, U_2 \dots U_{j-1}, F_j\}$ we first condition on all states of the system that have $(j-1)$ or more busy servers. For a given state we sequentially check the status of a list of servers until the first available server is found. If m out of a total of N servers are busy, the probability that the first randomly chosen $(j-1)$ servers are busy is a standard combinatorial problem and is

$$\binom{m}{j-1} / \binom{N}{j-1}$$

The conditional probability that the j th server is available is

$$(N-m)/N - (j-1).$$

Thus,

$$\begin{aligned} P(U_1, U_2 \dots U_{j-1}, F_j) &= \sum_{m=j-1}^{N-1} P(U_1, U_2 \dots U_{j-1}, F_j | S_m) P(S_m) \\ &= \sum_{m=j-1}^{N-1} \left[\binom{m}{j-1} / \binom{N}{j-1} \right] [N-m/N - (j-1)] P(S_m) \end{aligned} \quad (9)$$

where

$S_m \equiv$ exactly m servers are busy in the $M/M/N/0$ queuing system with bulk arrivals.

Unlike Larson's models, $P(S_m)$ does not take on a simple closed form expression. It is determined by solving the steady state queuing equations that are presented in Appendix II. Once $P(S_m)$ is calculated, we use equations (8) and (9) to find the first correction factor, which is

$$Q(N, j-1, 1) = \sum_{m=j-1}^{N-1} \frac{\binom{m}{j-1}}{\binom{N}{j-1}} \times \frac{N-m}{N-(j-1)} \times \frac{P(S_m)}{[\rho^{j-1}(1-\rho)]}. \quad (10)$$

The procedure for finding the second correction factor also involves conditioning on the state space, S_m , and carrying on a sequential check of server status. If m out of N servers are busy, the probability that only one server out of the first $(j-1)$ servers is busy is

$$\binom{m}{j-2} \binom{N-m}{1} / \binom{N}{j-1}.$$

Given that one of the first $(j-1)$ servers is busy, the probability that the server is the k th ranked server is $1/(j-1)$, and is not dependent on k . Therefore, the probability that only the k th server out of the first $(j-1)$ servers is busy is

$$\binom{m}{j-2} \binom{N-m}{1} / \binom{N}{j-1} (j-1).$$

The conditional probability that the j th server is available is

$$(N - m - 1)/N - (j - 1)$$

and the analog of equation (9) becomes

$$\begin{aligned} P(U_1, \dots, U_{k-1}, F_k, U_{k+1}, \dots, U_{j-1}, F_j) \\ = \sum_{m=j-2}^{N-1} \frac{\binom{m}{j-2}}{\binom{N}{j-1}} \frac{(N-m)}{(j-1)} \frac{N-m-1}{N-(j-1)} \times P(S_m). \end{aligned} \quad (11)$$

Equation (11) and the second component of equation (8) are used to find the second correction factor, which is

$$Q(N, j-1, 2) = \sum_{m=j-2}^{N-1} \frac{\binom{m}{j-2} (N-m)(N-m-1)}{\binom{N}{j-1} (j-1)(N-(j-1))} P(S_m) / \rho^{j-2} (1-\rho)^2 \quad (12)$$

and is independent of k . These correction factors provide the basis for calculating the average workload of unit i , ρ_i , as follows:

Let R_i = effective rate at which unit i is assigned to calls, given unit i is idle. By steady state arguments, albeit non-Markovian,

$$\rho_i = R_i / (\mu + R_i). \quad (13)$$

If we define R'_i to be the (unconditional) rate per unit time at which unit i is assigned to calls then

$$R'_i = R_i (1 - \rho_i). \quad (14)$$

An exact expression for R'_i can be written as a weighted sum of the call rates of the atoms for which unit i is ranked 1st, 2nd, 3rd, etc.

Let $G_i^k \equiv$ set of atoms for which unit i is the k th-ranked unit.

f_{mp} = fraction of the total Type p call rate that is generated by atom m ,

$W_i(W'_i)$ = event that unit i is (not) working,

n_{mj} = identification number of the j th preferred response unit for atom m .

Then for the *exact* model

$$\begin{aligned} R'_i = & \sum_{m \in G_i^1} (\lambda_1 f_{m1} + \lambda_2 f_{m2}) P(W'_i) + \sum_{m \in G_i^2} (\lambda_1 f_{m1} + \lambda_2 f_{m2}) P(W_{n_{m1}} W'_i) \\ & + \sum_{m \in G_i^2} \lambda_2 f_{m2} P(W'_{n_{m1}} W'_i) + \sum_{m \in G_i^3} (\lambda_1 f_{m1} + \lambda_2 f_{m2}) P(W_{n_{m1}} W_{n_{m2}} W'_i) \\ & + \sum_{m \in G_i^3} \lambda_2 f_{m2} [P(W'_{n_{m1}} W_{n_{m2}} W'_i) + P(W_{n_{m1}} W'_{n_{m2}} W'_i)] + \dots \\ & + \sum_{m \in G_i^N} (\lambda_1 f_{m1} + \lambda_2 f_{m2}) P(W_{n_{m1}} \dots W_{n_{m(N-1)}} W'_i) \\ & + \sum_{m \in G_i^N} \lambda_2 f_{m2} \sum_{k=1}^{N-1} P(W_{n_{m1}} \dots W_{n_{m(k-1)}} W'_{n_{mk}} W_{n_{m(k+1)}} \dots W_{n_{mN}} W'_i). \end{aligned} \quad (15)$$

Type 1 and Type 2 calls from each atom contribute to R'_i whenever unit i is the highest ranked available unit. Type 2 calls from each atom also contribute when unit i is the second highest ranked available unit. To approximate this expression, we estimate the probabilities as products of the individual utilization factors and the previously noted correction factors. (See equations (6) and (7).) We use the approximate version of equation (15) for R_i together with equations (13) and (14) to obtain an expression for ρ_i . The result is N simultaneous nonlinear equations that are solved iteratively, just as the 2^N linear state equations are solved in the exact model. An initial estimate of ρ_i is the average utilization as calculated from the basic $M/M/N/0$ queuing model. At the n th iteration, we obtain a new estimate $\hat{\rho}_i(n)$ for that unit's average utilization. Because the correction factor is not exact, at the end of each iteration the average of the $\hat{\rho}_i(n)$ may not equal the true region-wide average utilization. To insure that the average utilization is held constant, we normalize the individual unit workloads by multiplying each $\hat{\rho}_i(n)$ by the same constant.

The fraction of Type 1 dispatches that send the j th ranked unit to atom m requires weighting equation (6) by the atom's relative call rate. Let n_{mj} = the identification number of the unit ranked j th with respect to atom m .

$$q_{n_{mj},m1} = f_{m1}(1 - \rho_{n_{m1}}) \quad \text{for } j = 1,$$

$$q_{n_{mj},m1} = f_{m1} Q(N, j-1, 1) \left(\prod_{l=1}^{j-1} \rho_{n_{ml}} \right) (1 - \rho_{n_{mj}}) \quad \text{for } j \geq 2. \quad (16)$$

We similarly weight equation (7) to obtain the fraction of Type 2 dispatches that send the k th and j th ($j > k$) ranked units to atom m .

$$q_{(n_{mk}, n_{mj}), m2} = f_{m2} Q(N, j-1, 2) \left[\prod_{\substack{l=1 \\ l \neq k}}^{j-1} \rho_{n_{ml}} \right] (1 - \rho_{n_{mk}})(1 - \rho_{n_{mj}}) \quad \text{for } j > k. \quad (17)$$

The fraction of Type 2 dispatches that send just the j th ranked unit is given by equation (18).

$$q_{n_{mj}, m2} = f_{m2} Q(N, N-1, 1) \left(\prod_{l \neq j} \rho_{n_{ml}} \right) (1 - \rho_{n_{mj}}) \quad \text{for all } j. \quad (18)$$

The j th ranked unit will be the only unit dispatched to a Type 2 call only if all other units are busy. The server selection process involved in sending just the j th ranked unit to a Type 2 call could be viewed equivalently as checking all of the other servers and finding them unavailable for dispatch and then checking the status of the j th ranked unit and finding it available to be dispatched. As a result we use the Type 1 correction factor, $Q(N, N-1, 1)$, in equation (18); this corresponds to checking all of the servers and finding only the last one available to be dispatched. These statistics are then normalized as described in the exact model to insure that they add up to one.

5. The Calculation of Travel Time Statistics

We limit our discussion to the statistics unique to the multiple dispatch model, which were listed in §1. All of the statistics are calculated by taking the appropriate weighted sums of the dispatch statistics.

1. Region-wide mean travel time to Type 2 calls

$$\bar{T}_2 = \sum_{m=1}^M \left[\left[\sum_{j=i+1}^N \sum_{i=1}^{N-1} q_{(i,j)m2}(t_{im} + t_{jm}) \right] + \sum_{i=1}^N q_{im2}t_{im} \right]. \quad (19)$$

t_{im} = the average travel time from beat i (where unit i is on patrol) to atom m .

This statistic aggregates and averages the travel time of all units dispatched to Type 2 calls.

In the following statistics we separate the travel times of the two units dispatched to a Type 2 call. We estimate separately the travel time of the highest ranked available (primary) unit and the second highest ranked (backup) available unit. For the primary unit statistic, those Type 2 calls for which only one unit is dispatched are also included. Although unit i is ranked higher than unit j with regard to atom m , under most dispatch policies this does not mean that unit i is always closer. We therefore also estimate the average travel time of the first-arriving and second-arriving units when two units are dispatched. Let $L_m^i \equiv$ set of all units that are ranked lower than unit i with respect to atom m .

2. Mean Travel Time of the Primary Unit to a Type 2 Call

$$\sum_{m=1}^M \sum_{i=1}^N t_{im} \left[q_{im2} + \sum_{j \in L_m^i} q_{(i,j)m2} \right]. \quad (20)$$

3. Mean Travel Time of the Backup Unit to a Type 2 Call

$$\sum_{m=1}^M \sum_{i=1}^N \sum_{j \in L_m^i} q_{(i,k)m2} t_{jm}. \quad (21)$$

4. Mean Travel Time of the First-Arriving Unit

$$\bar{T}_C = \frac{\sum_{m=1}^M \sum_{j=i+1}^N \sum_{i=1}^{N-1} \sum_{r \in \text{beat } j} \sum_{k \in \text{beat } i} q_{(i,j)m2} l_{jr} l_{ik} \min[\tau_{rm}, \tau_{km}]}{D}. \quad (22)$$

5. Mean Travel Time of the Second-Arriving Unit

$$\bar{T}_F = \frac{\sum_{m=1}^M \sum_{j=i+1}^N \sum_{i=1}^{N-1} \sum_{r \in \text{beat } j} \sum_{k \in \text{beat } i} q_{(i,j)m2} l_{jr} l_{ik} \max[\tau_{rm}, \tau_{km}]}{D}. \quad (23)$$

where

$$D = \sum_{m=1}^M \sum_{j=i+1}^N \sum_{i=1}^{N-1} q_{(i,j)m2}.$$

The denominator is needed in the last two expressions because these travel time statistics include only dispatches involving two units.

The difference between equations (22) and (23) is the mean delay between the first-arriving and second-arriving units. Not all units, however, are equally likely to be the first-arriving unit at a Type 2 call, nor is the average wait for the second-arriving unit the same for each unit arriving first. The frequency with which unit i arrives first and is exposed to danger is just the proportion of all Type 2 dispatches in which its

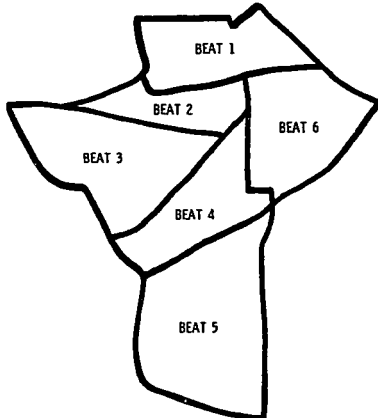


FIGURE 1. Sam Sector in New Haven, Connecticut.

travel time is the shorter of the two travel times. To find the average wait of unit i for the second-arriving unit, we condition on unit i arriving first at the scene and take a probabilistically weighted sum of $(\tau_{km} - \tau_{rm})$, the difference between the travel times to atom m of unit i traveling from atom k and unit j traveling from atom r . We also group these differences to find the probability distribution of the wait.

6. An Application to a Police Sector in New Haven, Connecticut

The data for this example are drawn largely from the New Haven Department of Police Services (Sam sector), which is described in a case study involving the application of the basic hypercube model [4]. The data were modified to generate Type 2 calls by assuming each atom's workload consisted of 80% Type 1 calls and 20% Type 2 calls. The dispatch strategy that was used was the modified center of mass strategy (MCM) described in [16]. Figure 1 is a map of Sam sector with each district's boundaries clearly delineated. Table 1 contains the correction factors for the highest-ranked and second-highest ranked available units (see equations (10) and (12)), which are used in the approximate model). In Tables 2, 3, and 4, we present estimates of workloads and travel times as calculated by the exact and approximate models.

We begin with a discussion of Sam sector's performance measures as calculated by the exact model (top line of the pairs in each table), and then proceed to compare them to the approximate estimates (second line). The mean travel time to a call in Sam sector is 5 minutes (Table 2). The mean travel time to Type 1 calls and the mean travel

TABLE 1
Correction Factors for Six-Unit Example

Correction Factors						
Rank $[j]$	1	2	3	4	5	6
First-Available $[Q(6, j - 1, 1)]$	1	0.853	0.809	0.838	0.936	1.114
Second-Available $[Q(6, j - 1, 2)]$	NA	1.182	0.909	0.772	0.716	0.715

TABLE 2

Sam Sector: Aggregate Travel Time Statistics
Mean Travel Time Statistics for Sector (Minutes)

	Overall Mean	Call Type 1	Call Type 2					Delay Between Units
			Combined	Primary Unit	Backup Unit	First- Arriving Unit	Second- Arriving Unit	
Exact	5.02	4.70	5.73	4.70	6.80	3.83	7.16	3.32
Approximate	4.98	4.68	5.65	4.64	6.76	3.78	7.11	3.33
Percentage Difference	0.8%	0.4%	1.4%	1.3%	0.6%	1.3%	0.6%	0.3%

TABLE 3

Sam Sector: Patrol Unit Statistics

Patrol Unit Number	Model ^a	Workload ^b	Overall	Call Type 1	Mean Travel Time (Minutes)			Call Type 2 Fraction of Time Unit <i>i</i> Arrives 1st
					Call Type 2			
					Combined ^c	First to Arrive ^d	Delay Until 2nd Arrival	
1	E	0.548	5.08	4.76	5.81	3.71	3.22	0.154
	A	.552	5.05	4.75	5.72	3.71	3.28	0.157
2	E	0.591	4.40	4.16	4.91	3.59	3.14	0.208
	A	0.601	4.40	4.17	4.87	3.60	3.22	0.206
3	E	0.578	4.52	4.22	5.24	3.70	3.41	0.187
	A	0.584	4.51	4.22	5.19	3.70	3.45	0.185
4	E	0.597	4.83	4.61	5.33	3.97	3.07	0.187
	A	0.604	4.83	4.62	5.30	3.97	3.10	0.186
5	E	0.462	6.56	6.05	7.61	4.19	4.17	0.106
	A	0.435	6.40	5.90	7.44	4.16	4.19	0.111
6	E	0.546	5.15	4.80	5.93	4.01	3.29	0.157
	A	0.546	5.11	4.77	5.85	3.98	3.30	0.156
Average	0.554							
Average% Difference	1.8%	0.6%	1.2%	0.7%	0.3%	1.2%	1.4%	
Maximum% Difference	6.2%	2.5%	2.2%	2.4%	0.8%	2.5%	4.5%	

Notes:

^aE = Exact model.

A = approximate model.

^bWorkload = average fraction of time busy servicing calls.^cCombined = Mean Travel Time of unit *i* to Type 2 calls.^dFirst to Arrive = Mean Travel Time of unit *i* to Type 2 call when it is the closer unit.

TABLE 4

Sam Sector: Beat Statistics

Beat Number	Model ^a	Mean Travel Time (Minutes)		
		Overall	Call Type 1	Call Type 2
1	E	5.14	4.81	5.87
	A	5.10	4.78	5.79
2	E	4.35	4.06	4.98
	A	4.34	4.07	4.93
3	E	5.30	4.85	5.97
	A	5.17	4.84	5.89
4	E	4.49	4.28	4.95
	A	4.47	4.28	4.90
5	E	6.14	5.63	7.24
	A	6.01	5.53	7.07
6	E	5.11	4.79	5.80
	A	5.08	4.79	5.73
Average Percentage Difference		0.8%	0.5%	1.4%
Maximum Percentage Difference		2.1%	1.7%	2.3%

Note:

^aE = exact model.

A = approximate model.

time of the primary unit dispatched to a Type 2 call are 6% lower than this (4.7 minutes). The backup unit assigned to a Type 2 call has a mean travel time of 5.7 minutes. More significantly, if we consider the perspective of the citizen caller, the first-arriving unit at a Type 2 call (when two units are dispatched) will reach the scene in four minutes, 20% less than the overall average. That unit will have to wait over three minutes before a second unit arrives at the scene.

The workload (fraction of time busy) of the units (Table 3) ranges from a low of 0.462 (unit 5) to a high of 0.597 (unit 4), which is 30% higher than unit 5's. Travel times by unit range from a low of 4.4 minutes (unit 2) to 50% higher for unit 5.

The mean travel time to different beats (a combination of atoms) shows a similar range of values. It is interesting to note that the difference between the mean travel time to a Type 1 call and a Type 2 call varies significantly from beat to beat. In beat 5, which is at the southern tip of the sector (see Figure 1), Type 2 calls experience travel times that average 1.6 minutes longer than Type 1 calls. In contrast, beat 4, which is more centrally located, has a difference of only 0.73 minutes.

Another important set of statistics describes the arrival pattern at a Type 2 call. Unit two is the first unit to arrive at 21% of the sector's Type 2 calls. When unit two is the first to arrive, its average travel time is 3.6 minutes and it has to wait an average of 3.1 minutes until help arrives. In contrast, unit five is the first unit to arrive at only 11% of

the sector's Type 2 calls but will wait an average of 4.2 minutes for help.

The relative frequency with which unit i is the first unit to arrive at a Type 2 call and the average wait for help are surrogate measures of the frequency and degree of danger that the single officer is exposed to. Consequently, the inequities in these statistics that appear in this example would seem to be of far greater concern than inequities resulting in one unit being busier than the other. This example highlights what should be a new concern for the police manager constructing beats for one-officer units: How to construct beats so that exposure at Type 2 calls is equitably distributed among the units.

Comparison of Exact and Approximate Models

The estimates of the various performance measures from the two models are very close. Overall, the travel time and workload estimates of the two models differ by an average of less than 2%. The maximum difference between the two models' workload estimates is 6.2%. The approximate model underestimated the workload of the least busy unit; this difference was balanced by several smaller overestimates of the busier units' workloads. Differences in the travel time estimates generally do not exceed 3%. The approximate model therefore provides sufficiently accurate estimates for planning purposes. This is especially important as the size of the region of interest grows, since it quickly becomes infeasible to use the exact model. In this example, which involved only six units, the exact model required 157.5 cpu seconds to run, while the approximate model required only 14.5 seconds (less than one-tenth of the exact model).⁵

7. Future Research

The models presented here are intended to serve both as local police planner's tools and as catalysts for future research into the trade-offs between one- and two-officer patrol units. Although these models build on the hypercube queuing model, which has been validated, there is an obvious need to validate the new models. The focus of any validation should be on the new travel time statistics calculated by these models. One prerequisite for such a validation is that the police department's records contain the travel time of each of the one-officer units sent to a Type 2 call, and that it be possible to easily link up the travel time records of pairs of units sent to the same call. The linked records would be used to determine the proportion of time unit i arrives first and the average time unit i waits for help to arrive. (See Chelst [9] for a comparable study of dual response to emergency medical calls.)

The most straightforward expansion of the model's capabilities is to allow Type 1 calls to be queued. This can be done in much the same way that Larson's hypercube model [17] and approximate analog [18] allow for a queue. A basic queuing model without identifiable servers would be used to calculate the probability of a queue, $P(Q)$, and the probability that n servers are busy $P(S_n)$. Once these probabilities are obtained, both the exact and approximate models proceed as before.

In contrast, the addition of a queue for Type 2 calls is a complex problem. When no Type 2 queue is allowed and servers are not identified, the system is indistinguishable

⁵All computations were done on Wayne State University's Amdahl V/6.

from a bulk arrival queuing model. Whether two customers arrive simultaneously or one customer arrives requiring two servers, the number of busy servers increases by two. A Type 2 call that is queued, however, cannot be treated the same as two customers that are queued [10]. In the latter case, as busy servers complete service, each customer is treated individually and two queued customers might even be served by the same server. On the other hand, the Type 2 call needs two servers working at least part of the time together. What happens when a Type 2 call is at the head of the queue and one server becomes available? Is the call held until another server becomes available or is service begun immediately? Either choice poses modeling problems and will require at the very least that the state space differentiate in the *queue* between Type 1 and Type 2 calls.

A more fundamental model expansion, which was discussed earlier, is to allow service of Type 2 calls to differ substantially from Type 1 calls. One possibility would allow both servers begin and end service simultaneously. Another would allow the mean service time for Type 2 calls to be different from Type 1 calls.

Using Models to Analyze the Merger of Emergency Services

A number of smaller cities around the country have merged their police, fire and EMS services into public safety departments. In the face of growing municipal fiscal crises, more city officials are likely to consider this alternative. Except for limited surveys of existing public safety departments [12], there is little research to guide a city administrator in evaluating the concept in his/her locale or provide advice as to how to implement the concept. Although a primary motivation for merging emergency services is economic, an obvious concern of any administrator contemplating a merger is its impact on system performance.

Existing models of police and fire deployment can be used to evaluate the impact on performance of traditional cost saving options of reducing manpower and/or closing fire stations. Two modifications of the models described in this paper would provide the city manager with a tool for evaluating the impact of a merger on performance.

- Introduce separate dispatch lists in order to model the dispatch of different types of units to the same emergency.

- Allow for the dispatch of more than two units to model the response to a fire.

With these modifications in place, the model can be used to estimate a variety of response time statistics.

- the response time of the first officer to arrive at an emergency.

- the average time for a patrol unit and an ambulance to respond to a medical emergency.

- the average time to get a full complement of officers to a fire to man a fire engine.

In addition to addressing the basic question of whether or not to merge, the model can help explore other issues that will arise in a merger. A merger is likely to be phased in over the period of several years, as officers leave their respective departments and personnel are trained to handle multiple roles.

- How will the system operate during the phasing in period?

- What is the minimum number of public safety officers that must be deployed before the city can begin to take advantage of their eclectic training?

An important operational decision for a merged system is

How many officers should be kept in the station to travel with the fire engine or ladder?

If only a few officers are left to travel with the equipment, a large number of public safety units will also have to be dispatched to a fire scene. The model can be used to explore this tradeoff between more officers in the station and response times.

Without a model such as this to predict system performance and additional models to project cost savings and plan manpower changes, a city manager cannot help but view a merger as a political quagmire on which even political angels fear to tread.

Conclusion

In this paper we have presented exact and approximate models of an emergency service in which two identical units are often dispatched to one call. These models can estimate new performance measures that are relevant to this dispatch environment. We have applied these models to an example and highlighted issues that need to be addressed when deploying one-officer patrol units. Future development will concentrate on the approximate version, since as assumptions are relaxed, the state space of the exact model expands to become computationally infeasible for even medium sized emergency service departments.

Appendix I

Table of Symbols

B_i	A vertex contained in C_N which is represented as a vector of b_n 's.
b_n	The n th binary digit in B_i which represents the status of the n th unit. If $b_n = 1$ the server is busy and if $b_n = 0$ it is available.
C_N	Vertices of N -dimensional unit hypercube in the positive orthant.
d_{ij}	Hamming distance between two vertices B_i and B_j .
D_m^i	Set of all states in which, with respect to atom m , unit i is the highest ranked available unit.
$D_m^{i,j}$	Set of all states in which, with respect to atom m , units i and j are the highest ranked available units.
F_j	Event that the j th server selected is free.
$f_{mp}^{k,i}$	Fraction of call type p region-wide workload generated from atom m .
G_i^k	Set of geographical atoms for which unit i is the k th preferred dispatch alternative.
H_n	n th hyperplane from the origin. All states, B_i , in a hyperplane have the same $w\{B_i\}$.
L_m^i	Set of all units which are ranked lower than unit i with respect to atom m .
M	Total number of geographical atoms.
N	Total number of response units.
n_{mj}	Identification number of the j th preferred response unit for atom m .
$P(S_k)$	Probability that exactly k servers are busy.
$Q(N, j, 1)$	Correction factor for computing the probability that the $j + 1$ st ranked server is the first available server given a total of N servers.
$Q(N, j, 2)$	Correction factor for computing the probability that the $j + 2$ nd ranked server is the second available server.
q_{im1}	Fraction of Type 1 dispatches which send unit i to atom m .
$q_{(i, j)m2}$	Fraction of Type 2 dispatches which send units i and j to atom m .
q_{im2}	Fraction of Type 2 dispatches which send only unit i to atom j .

R_i	Effective rate at which unit i is assigned to calls given that it is idle.
S_k	State of the $M/M/M/0$ queuing system indicating that exactly k servers are busy.
t_{im}	Mean travel time from beat i to atom m .
U_j	Event that the j th server selected is unavailable.
$w\{B_i\}$	Weight of vertex B_i , $\sum_{n=1}^N b_n$.
W_i	Event that unit i is working.
W'_i	Event that unit i is not working.
λ_k	Average number of type k calls for service generated per hour from within the entire region of interest.
$\lambda_{ij}^{(k)}$	Infinitesimal mean rate of type k transitions from state i to state j , given the system is in state i .
μ^{-1}	Average service time of a unit at a call.
ρ_n	Workload of unit n (fraction of time busy servicing calls).
l_{nm}	Probability that response unit n is located in atom m while available.
τ_{im}	Mean travel time from atom i to atom m .

Appendix II

Balance Equations for Bulk Arrivals, Multi-Server, Zero-Capacity Queue.

Let $\lambda = \lambda_1 + \lambda_2$ and $N =$ number of servers:

$$\lambda P_0 = \mu P_1,$$

$$(\lambda + \mu)P_1 = \lambda_1 P_0 + 2\mu P_2,$$

$$(\lambda + n\mu)P_n = \lambda_2 P_{n-2} + \lambda_1 P_{n-1} + (n+1)\mu P_{n+1} \quad \text{for } n = 2, 3, \dots, N-1,$$

$$N\mu P_N = \lambda_2 P_{N-2} + \lambda P_{N-1}.$$

In addition $\sum_{i=0}^N P_i = 1$. This last equality is used to replace any one of the $N+1$ steady state equations.⁶

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